

Employee Attrition Prediction – Project Summary

Project Objective

The objective of this project was to identify employees who may be at risk of leaving the company so that the Human Resources (HR) department can take preventive actions before valuable employees resign. Employee attrition can increase recruitment costs, reduce productivity, and affect overall business performance. By analysing employee information, this project aims to support better workforce planning and retention strategies.

Key Findings

The analysis showed that employee attrition is influenced by several workplace factors rather than a single reason. Employees working in certain job roles, especially Laboratory Technicians and Sales Representatives, showed a higher likelihood of leaving the organization. Employees who frequently worked overtime or travelled regularly for business also appeared to have a greater risk of attrition. Career-related factors, such as total working years and the time since the last promotion, also played an important role.

The analysis further indicated that salary alone does not fully explain why employees leave. Workload, job responsibilities, career growth opportunities, and work-life balance all contribute to employee retention.

Business Recommendations

Based on these findings, the HR department should focus on reducing excessive overtime, improving work-life balance, and creating targeted retention programs for employees working in high-risk job roles. Regular career development discussions, recognition programs, and promotion opportunities may also help reduce employee turnover.

This predictive model should be used as a decision-support tool to help HR identify employees who may require additional attention. However, final decisions should always consider individual employee circumstances and organizational factors that may not be included in the available data.

Conclusion

This project demonstrates how data analysis can support HR in making proactive decisions to improve employee satisfaction and reduce attrition. By identifying employees who may be at risk of leaving, organizations can take timely actions that improve employee retention, reduce hiring costs, and strengthen overall organizational performance.